

Gradient Descent with Polyak-Lojasiewicz Condition: Algorithm and Convergence Analysis

Optimization Expert

June 12, 2025

1 Algorithm Description

The algorithm iteratively updates parameters by computing gradients on randomly sampled data points and taking steps in the negative gradient direction with a fixed step size. It is designed for smooth functions satisfying the Polyak-Lojasiewicz condition.

1.1 Mathematical Formulation

1.1.1 Objective

Minimize $f(\mathbf{w})$, where f is a smooth function satisfying the Polyak-Lojasiewicz (PL) condition:

$$f(\mathbf{w}) - f(\mathbf{w}^*) \leq \frac{1}{\mu} \|\nabla f(\mathbf{w})\|^2$$

for some $\mu > 0$ and $\mathbf{w}^*$ being an optimal solution.

1.1.2 Update Rules and Equations

Given:

- $\mathbf{w}_t$: Parameters at iteration t
- η : Fixed step size (hyperparameter)
- $\mathcal{D}$: Dataset
- B : Minibatch size (hyperparameter)

1. Gradient Computation on Minibatch:

$$\nabla f(\mathbf{w}_t; \mathcal{B}_t) = \frac{1}{B} \sum_{i \in \mathcal{B}_t} \nabla f_i(\mathbf{w}_t)$$

where $\mathcal{B}_t$ is a minibatch of size B randomly sampled from $\mathcal{D}$, and f_i is the loss for the i -th data point.

2. Parameter Update:

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla f(\mathbf{w}_t; \mathcal{B}_t)$$

1.1.3 Hyperparameters and Their Roles

- η (Step Size): Controls how large each step in the negative gradient direction is. Too large can lead to divergence, too small can lead to slow convergence.
- B (Minibatch Size): Balances between the stability of gradient estimates (larger B) and computational efficiency (smaller B).

2 Pseudo Code and Dry Run

2.1 Pseudocode

Algorithm 1 Gradient Descent with Polyak-Lojasiewicz Condition (GD-PL)

Input:

Dataset $\mathcal{D}$
Initial parameters $\mathbf{w}_0$
Step size η
Minibatch size B
Maximum iterations T
Tolerance ϵ (optional)

Output:

Optimized parameters $\mathbf{w}_T$
Initialize $t = 0$
while $t < T$ [and $\|\nabla f(\mathbf{w}_t)\| > \epsilon$ if using tolerance] **do**
 Sample a minibatch $\mathcal{B}_t$ of size B from $\mathcal{D}$
 Compute the gradient on the minibatch:
 $\nabla f(\mathbf{w}_t; \mathcal{B}_t) = \frac{1}{B} \sum_{i \in \mathcal{B}_t} \nabla f_i(\mathbf{w}_t)$
 Update parameters:
 $\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla f(\mathbf{w}_t; \mathcal{B}_t)$
 $t \leftarrow t + 1$
end while
return $\mathbf{w}_T$

2.2 Dry Run Example

2.2.1 Function

$$f(\mathbf{w}) = (\mathbf{w} - 3)^2$$

2.2.2 Initialization

$$\mathbf{w}_0 = 0$$

2.2.3 Step Size (η)

$$0.1$$

2.2.4 Minibatch Size (B)

1 (deterministic, as f is not data-dependent in the traditional sense but for demonstration)

2.2.5 Iterations

Iteration 1

- **Gradient Computation:**

$$\nabla f(\mathbf{w}_0) = 2(\mathbf{w}_0 - 3) = 2(0 - 3) = -6$$

- **Parameter Update:**

$$\mathbf{w}_1 = \mathbf{w}_0 - \eta \cdot \nabla f(\mathbf{w}_0) = 0 - 0.1 \cdot (-6) = 0.6$$

- **Objective Value:**

$$f(\mathbf{w}_1) = (0.6 - 3)^2 = (-2.4)^2 = 5.76$$

Iteration 2

- **Gradient Computation:**

$$\nabla f(\mathbf{w}_1) = 2(\mathbf{w}_1 - 3) = 2(0.6 - 3) = -4.8$$

- **Parameter Update:**

$$\mathbf{w}_2 = \mathbf{w}_1 - \eta \cdot \nabla f(\mathbf{w}_1) = 0.6 - 0.1 \cdot (-4.8) = 0.6 + 0.48 = 1.08$$

- **Objective Value:**

$$f(\mathbf{w}_2) = (1.08 - 3)^2 = (-1.92)^2 = 3.6864$$

Iteration 3

- **Gradient Computation:**

$$\nabla f(\mathbf{w}_2) = 2(\mathbf{w}_2 - 3) = 2(1.08 - 3) = -3.84$$

- **Parameter Update:**

$$\mathbf{w}_3 = \mathbf{w}_2 - \eta \cdot \nabla f(\mathbf{w}_2) = 1.08 - 0.1 \cdot (-3.84) = 1.08 + 0.384 = 1.464$$

- **Objective Value:**

$$f(\mathbf{w}_3) = (1.464 - 3)^2 = (-1.536)^2 = 2.356096$$

2.2.6 Summary Table

Iteration	$\mathbf{w}$	Gradient	$f(\mathbf{w})$
0	0	-6	9
1	0.6	-4.8	5.76
2	1.08	-3.84	3.6864
3	1.464	-3.072	2.356096

Table 1: Summary of Dry Run Iterations

3 Convergence Theory

3.1 Assumptions

3.1.1 Formal Statement of Assumptions

1. Polyak-Lojasiewicz (PL) Condition:

- **Definition:** A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies the Polyak-Lojasiewicz condition with constant $\mu > 0$ if for all $x \in \mathbb{R}^d$,

$$\frac{1}{2} \|\nabla f(x)\|^2 \geq \mu(f(x) - f^*)$$

where $f^* = \min_{x \in \mathbb{R}^d} f(x)$.

- **Parameter Constraint:** $\mu > 0$.
- **Function Property:** f is bounded below.

2. Smoothness Condition:

- **Definition:** A differentiable function f is L -smooth if for all $x, y \in \mathbb{R}^d$,

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$$

- **Parameter Constraint:** $L > 0$.
- **Function Property:** f is continuously differentiable.

3. Algorithmic Assumptions:

- **Fixed Step Size:** $\alpha > 0$.
- **Random Sampling:** At each iteration, a data point is sampled uniformly at random from a finite set $\{x_1, x_2, \dots, x_N\}$.

3.1.2 Summary of Assumptions Table

Assumption	Parameter Constraints	Function Properties
Polyak-Lojasiewicz	$\mu > 0$	Bounded Below
Smoothness	$L > 0$	Continuously Differentiable
Algorithmic	$\alpha > 0$	-

Table 2: Summary of Assumptions

3.2 Main Convergence Theorem

3.2.1 Theorem Statement

Theorem 1 (Convergence Rate): Under the Polyak-Lojasiewicz and smoothness assumptions, with a fixed step size $\alpha = \frac{1}{L}$, the expected function value of the parameters updated by the algorithm satisfies

$$\mathbb{E}[f(x_T)] - f^* = O\left(\left(1 + \frac{\mu}{L}\right)^{-T}\right)$$

where T is the number of iterations.

3.2.2 Proof Sketch

1. **Expected Gradient Norm:** Show $\mathbb{E}[\|\nabla f(x_t)\|^2] \geq 2\mu(\mathbb{E}[f(x_t)] - f^*)$ using PL condition.
2. **Smoothness Implies Progress:** Utilize L -smoothness to show $f(x_{t+1}) \leq f(x_t) - \frac{\alpha}{2}\|\nabla f(x_t)\|^2 + \frac{\alpha L}{2}\|\nabla f(x_t)\|^2$.
3. **Step Size Choice:** With $\alpha = \frac{1}{L}$, simplify to $f(x_{t+1}) \leq f(x_t) - \frac{1}{2L}\|\nabla f(x_t)\|^2$.
4. **Combine with PL:** Derive contraction $\mathbb{E}[f(x_{t+1})] - f^* \leq \left(1 + \frac{\mu}{L}\right)^{-1} (\mathbb{E}[f(x_t)] - f^*)$.
5. **Iterate Contractive Bound:** Over T iterations to achieve the stated rate.

3.2.3 Expected Reduction Properties

- **Per Iteration Reduction:** $\mathbb{E}[f(x_{t+1})] - f^* \leq \left(1 + \frac{\mu}{L}\right)^{-1} (\mathbb{E}[f(x_t)] - f^*)$
- **Special Case (Linear Convergence):** When μ/L is a constant, the algorithm exhibits linear convergence.

3.2.4 Special Case Behaviors

- **Purely Convex** ($\mu = 0$): The theory does not directly apply; expect slower convergence rates (e.g., $O(1/T)$ for Gradient Descent in convex settings).
- **Strongly Convex** ($\mu > 0$): Linear convergence as stated.

3.3 Theoretical Properties

3.3.1 Stability Analysis

- **Robustness to Noise**: The algorithm’s convergence is robust to stochastic gradient noise, with the noise variance affecting the constant in the convergence rate but not the rate itself.
- **Step Size Sensitivity**: Deviating from $\alpha = 1/L$ can lead to slower convergence or divergence if α is too large.

3.3.2 Sensitivity to Hyperparameters

Hyperparameter	Impact on Convergence
α (Step Size)	Too large: Divergence. Too small: Slower convergence.
μ (PL Constant)	Larger μ : Faster convergence.
L (Smoothness)	Larger L : Slower convergence for fixed α .

Table 3: Sensitivity to Hyperparameters

3.3.3 Comparison to Baseline Methods

- **Gradient Descent (GD) on Smooth Functions**: Similar convergence without PL condition but potentially slower (e.g., $O(1/T^2)$ with Nesterov acceleration in convex settings).
- **Stochastic Gradient Descent (SGD)**: This algorithm is a variant of SGD; the analysis here provides a tighter convergence bound under the PL condition.

3.4 Discussion

The developed convergence theory highlights the efficiency of the described algorithm under the Polyak-Lojasiewicz and smoothness conditions, offering linear convergence. This is particularly advantageous for functions where the PL condition holds, as it guarantees a faster convergence rate compared to standard convex optimization problems. The sensitivity analysis underscores the importance of appropriately choosing the step size α and highlights how the interplay between μ and L affects the convergence speed.

Future Directions:

- **Adaptive Step Sizing**: Investigate strategies to adapt α based on μ and L estimates.
- **Non-Smooth Objectives**: Extend the theory to non-smooth functions using subgradient methods or proximal operators.

4 Convergence Proofs

4.1 Preliminaries (Definitions and Lemmas)

4.1.1 Definitions

1. **Polyak-Lojasiewicz (PL) Condition**:

- Given a differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, it satisfies the PL condition with constant $\mu > 0$ if:

$$\frac{1}{2} \|\nabla f(x)\|^2 \geq \mu(f(x) - f^*) \quad \forall x \in \mathbb{R}^d$$

where $f^* = \min_{x \in \mathbb{R}^d} f(x)$.

2. Smoothness Condition:

- f is L -smooth if:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \quad \forall x, y \in \mathbb{R}^d$$

3. Algorithm:

- Update rule: $x_{t+1} = x_t - \alpha \nabla f_{i_t}(x_t)$, where i_t is uniformly random over $\{1, 2, \dots, N\}$, and $\alpha = \frac{1}{L}$.

4.1.2 Lemmas

1. Expected Gradient Norm (PL Implication):

$$\mathbb{E}[\|\nabla f(x_t)\|^2] \geq 2\mu(\mathbb{E}[f(x_t)] - f^*)$$

2. Smoothness Progress:

- For L -smooth f ,

$$f(y) \leq f(x) + \nabla f(x)^T(y - x) + \frac{L}{2}\|y - x\|^2$$

4.2 Main Proof (Step-by-Step Derivation)

4.2.1 Step 1: Expected Gradient Norm via PL Condition

- **Given:** PL Condition
- **Show:** $\mathbb{E}[\|\nabla f(x_t)\|^2] \geq 2\mu(\mathbb{E}[f(x_t)] - f^*)$
- **Proof:**
 - By PL, $\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu(f(x) - f^*)$
 - Expectation over x_t : $\mathbb{E}[\frac{1}{2}\|\nabla f(x_t)\|^2] \geq \mu(\mathbb{E}[f(x_t)] - f^*)$
 - Multiply by 2: $\mathbb{E}[\|\nabla f(x_t)\|^2] \geq 2\mu(\mathbb{E}[f(x_t)] - f^*)$

□

4.2.2 Step 2: Smoothness Implies Progress

- **Given:** L -smoothness, Update Rule
- **Show:** $f(x_{t+1}) \leq f(x_t) - \frac{\alpha}{2}\|\nabla f(x_t)\|^2 + \frac{\alpha L}{2}\|\nabla f(x_t)\|^2$
- **Proof** (Using Smoothness Lemma with $y = x_t - \alpha \nabla f(x_t)$):

$$f(x_{t+1}) \leq f(x_t) - \alpha\|\nabla f(x_t)\|^2 + \frac{L\alpha^2}{2}\|\nabla f(x_t)\|^2$$

- Given $\alpha = \frac{1}{L}$:

$$\begin{aligned} f(x_{t+1}) &\leq f(x_t) - \frac{1}{L}\|\nabla f(x_t)\|^2 + \frac{1}{2L}\|\nabla f(x_t)\|^2 \\ &\Rightarrow f(x_{t+1}) \leq f(x_t) - \frac{1}{2L}\|\nabla f(x_t)\|^2 \end{aligned}$$

□

4.2.3 Step 3: Combine with PL for Contraction

- **Objective:** Derive $\mathbb{E}[f(x_{t+1})] - f^* \leq \left(1 + \frac{\mu}{L}\right)^{-1} (\mathbb{E}[f(x_t)] - f^*)$
- **Proof:**
 1. Take expectation of Step 2's result:

$$\mathbb{E}[f(x_{t+1})] \leq \mathbb{E}[f(x_t)] - \frac{1}{2L} \mathbb{E}[\|\nabla f(x_t)\|^2]$$

2. Apply Lemma 1:

$$\begin{aligned} \mathbb{E}[\|\nabla f(x_t)\|^2] &\geq 2\mu(\mathbb{E}[f(x_t)] - f^*) \\ \Rightarrow \mathbb{E}[f(x_{t+1})] &\leq \mathbb{E}[f(x_t)] - \frac{\mu}{L}(\mathbb{E}[f(x_t)] - f^*) \end{aligned}$$

3. Rearrange:

$$\begin{aligned} \mathbb{E}[f(x_{t+1})] - f^* &\leq \left(1 - \frac{\mu}{L}\right) (\mathbb{E}[f(x_t)] - f^*) \\ \Rightarrow \mathbb{E}[f(x_{t+1})] - f^* &\leq \left(1 + \frac{\mu}{L}\right)^{-1} (\mathbb{E}[f(x_t)] - f^*) \end{aligned}$$

□

4.2.4 Step 4: Iterate Contractive Bound

- **Objective:** Achieve $O\left(\left(1 + \frac{\mu}{L}\right)^{-T}\right)$
- **Proof (By Induction):**
 - Base Case ($T = 1$): Established in Step 3
 - Inductive Step: Assume true for T , then for $T + 1$:

$$\begin{aligned} \mathbb{E}[f(x_{T+1})] - f^* &\leq \left(1 + \frac{\mu}{L}\right)^{-1} (\mathbb{E}[f(x_T)] - f^*) \\ &\leq \left(1 + \frac{\mu}{L}\right)^{-1} \cdot \left(1 + \frac{\mu}{L}\right)^{-T} (\mathbb{E}[f(x_0)] - f^*) \\ &= \left(1 + \frac{\mu}{L}\right)^{-(T+1)} (\mathbb{E}[f(x_0)] - f^*) \end{aligned}$$

□

4.3 Rate Derivation (With Constants)

Given the contractive bound:

$$\mathbb{E}[f(x_T)] - f^* \leq \left(1 + \frac{\mu}{L}\right)^{-T} (\mathbb{E}[f(x_0)] - f^*)$$

- **Rate:** $O\left(\left(1 + \frac{\mu}{L}\right)^{-T}\right)$
- **Constants Dependence:**
 - **Linear Convergence Rate:** Dictated by $\frac{\mu}{L}$
 - **Initial Condition:** $\mathbb{E}[f(x_0)] - f^*$ affects the constant but not the rate

4.4 Special Cases

4.4.1 Purely Convex ($\mu = 0$)

- **Theory Limitation:** The derived rate does not apply
- **Expected Convergence:**
 - For Gradient Descent on convex functions, expected rate is $O\left(\frac{1}{T}\right)$, assuming bounded gradients or similar conditions for stochastic settings.

4.4.2 Strongly Convex ($\mu > 0$)

- **Convergence:**
 - **Linear Rate:** As stated, $O\left(\left(1 + \frac{\mu}{L}\right)^{-T}\right)$
 - **Implication:** Fast convergence guaranteed by the strong convexity and the PL condition.